UNIFORM VANISHING OF EXT FROM FINITE LOCALLY FREE MODULES INTO A BOUNDED DERIVED QUASI-COHERENT OBJECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Mayer-Vietoris for morphisms in the derived category and the induction principle for quasi-compact opens. invoked at proof lines 3428-3434. Glues the vanishing bound and reduces to the affine case, at the cost of increasing the bound by one at each gluing.
- On an affine scheme the derived category of quasi-coherent modules is equivalent to the derived category of modules, and for a complex with projective terms morphisms in the derived category are computed on the level of cohomology. invoked at proof lines 3437-3453. Turns the Ext group into Hom_A(P, H^n(M^bullet)).
- Hom out of a projective module is an exact functor. invoked at proof lines 3452-3453. Justifies the final equality with Hom into the cohomology module.

Source reference. Original lemma 08BW: sources/stacks-chow/tags.txt, source lines 10581–10581. Cited at proof lines 3430-3431 to glue the vanishing bound over unions; the reduction is inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Cited at proof lines 3433-3434 to reduce to the affine case; the affine case assembly is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 3438-3439 and 3449-3450 to represent E by M^bullet and translate the Hom; the computation is inside the counted span.

Source reference. Original lemma 064B: sources/stacks-chow/tags.txt, source lines 7860–7860. Cited at proof lines 3451-3452 for the second equality; the final vanishing argument is inside the counted span.

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Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $E \in D^b_{QCoh}(\mathcal{O}_X)$. There exists an integer $n_0 > 0$ such that $\mathrm{Ext}^n_{D(\mathcal{O}_X)}(\mathcal{E}, E) = 0$ for every finite locally free $\mathcal{O}_X$-module $\mathcal{E}$ and every $n \geq n_0$.*

Proof. Recall that $\mathrm{Ext}^n_{D(\mathcal{O}_X)}(\mathcal{E}, E) = \mathrm{Hom}_{D(\mathcal{O}_X)}(\mathcal{E}, E[n])$. We have Mayer-Vietoris for morphisms in the derived category, see Cohomology, Lemma 08BW. Thus if $X = U \cup V$ and the result of the lemma holds for $E|_U$, $E|_V$, and $E|_{U \cap V}$ for some bound n_0, then the result holds for E with bound $n_0 + 1$. Thus it suffices to prove the lemma when X is affine, see Cohomology of Schemes, Lemma 08DR.

Assume $X = \mathrm{Spec}(A)$ is affine. Choose a complex of A-modules $M^\bullet$ whose associated complex of quasi-coherent modules represents E, see Lemma 06Z0. Write $\mathcal{E} = \widetilde{P}$ for some A-module P. Since $\mathcal{E}$ is finite locally free, we see that P is a finite projective A-module. We have

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{O}_X)}(\mathcal{E}, E[n]) &= \mathrm{Hom}_{D(A)}(P, M^\bullet[n]) \\ &= \mathrm{Hom}_{K(A)}(P, M^\bullet[n]) \\ &= \mathrm{Hom}_A(P, H^n(M^\bullet)) \end{aligned}$$

The first equality by Lemma 06Z0, the second equality by Derived Categories, Lemma 064B, and the final equality because $\mathrm{Hom}_A(P, -)$ is an exact functor. As E and hence $M^\bullet$ is bounded we get zero for all sufficiently large n. $\square$